

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: pNeo Xp 5.3TK, pHyg Xp5.3 TK, about 10 kB.
Species Used	Mouse, embryonic stem cells		

Before the Pulse

Cell Growth Medium	20% Fetal Calf Serum, DMEM + amino acids, 2- mercaptoethanol, nucleosides. (GIBCO/ BRL, Sigma)	Growth Phase at Harvest	Log phase
		Pre-pulse Incubation	5 min., on ice

Wash Solution Phosphate Buffered Saline

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	4 to 10 °C	Cuvette Gap	0.4cm
Electroporation Medium*	Phosphate Buffered Saline	Voltage	0.28 kV
Cell Density	1 x 10 ⁷ (7) cells / ml	Field Strength	0.7 kV/cm
Volume of Cells	0.8 ml	Capacitor	500 µF
DNA Concentration	20 µg DNA/ pulse	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	Not given	Time Constant	5.8 to 6.4 msec
Volume of DNA	16 µl		

After the Pulse

Outgrowth Medium Not given

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	10 days
Selection Method or Assay Used	G418, GANC
Electroporation Efficiency	20 to 30 transfectants / µg DNA
Per Cent Survival	20 to 30%

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